

NYCU Introduction to Machine Learning, Homework 1

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Part. 1, Coding (60%):

(10%) Linear Regression Model - Closed-form Solution

1. (10%) Show the weights and intercepts of your linear model.

```
2025-10-06 03:50:41.492 | INFO | __main__:main:114 - LR_CF.weights=array([2.85501274, 1.01785863, 0.47202168, 0.19608925]), LR_CF.intercept=-33.6956
```

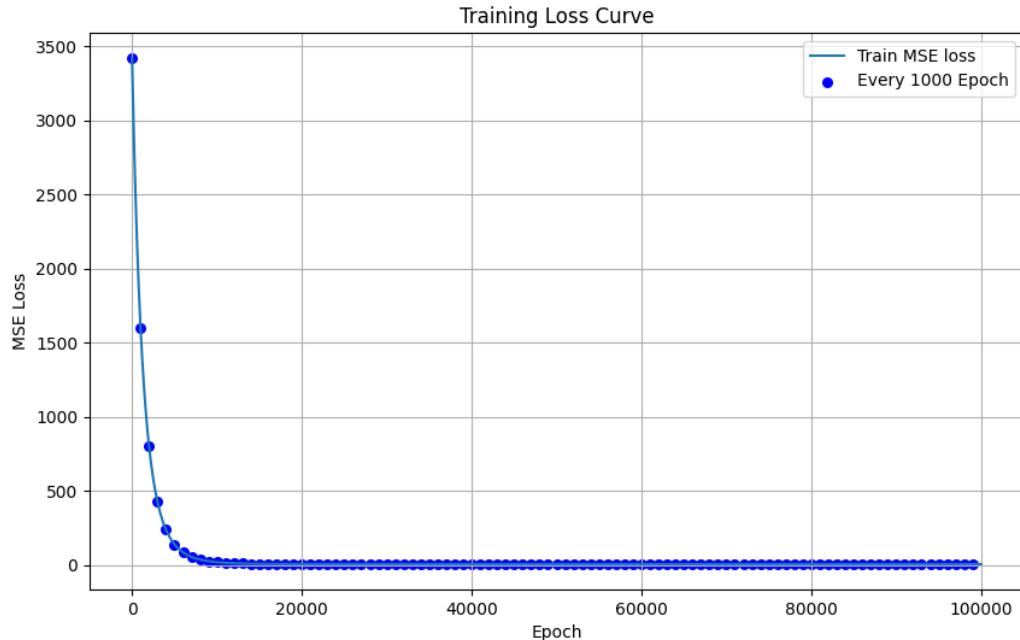
(40%) Linear Regression Model - Gradient Descent Solution

2. (10%)
 - Show the hyperparameters of your setting (e.g., learning rate, number of epochs, batch size, etc.).
 - Show the weights and intercepts of your linear model.

```
losses, lr_history = LR_GD.fit(  
    G_train_x, train_y, learning_rate=2e-4, epochs=100000  
)
```

```
2025-10-06 16:16:33.201 | INFO | __main__:main:134 - LR_GD.weights=array([ 7.39277106, 17.59488313, 0.80110683, 0.56401163]), LR_GD.intercept=55.2821
```

3. (10%) Plot the learning curve. (x-axis=epoch, y-axis=training loss)



4. (20%) Show your MSE.cf, MSE.gd, and error rate between your closed-form solution and the gradient descent solution.

```
2025-10-06 16:16:33.347 | INFO | __main__:main:164 - Prediction difference: 0.0038  
2025-10-06 16:16:33.347 | INFO | __main__:main:169 - mse_cf=4.2903, mse_gd=4.2904. Difference: 0.002%
```

(10%) Code Check and Verification

5. (10%) Lint the code and show the PyTest results.

```
(hw1) wuhsinze@HsinZewUWU hw1 % flake8 main.py
(hw1) wuhsinze@HsinZewUWU hw1 %

(hw1) wuhsinze@HsinZewUWU hw1 % pytest ./test_main.py -s
===== test session starts =====
platform darwin -- Python 3.13.7, pytest-8.4.2, pluggy-1.6.0
rootdir: /Users/wuhsinze/Desktop/三上/ML/HW1/hw1
collected 2 items

test_main.py (100, 1)
2025-10-06 16:15:47.194 | INFO | test_main:test_regression_cf:30 - model.weights=array([[3.]]), model.intercept=array([4.])
.(100, 1)
2025-10-06 16:15:47.194 | INFO | main:fit:81 - EPOCH 0, loss=30622.0606, LR=0.000200
2025-10-06 16:15:47.255 | INFO | main:fit:81 - EPOCH 10000, loss=0.0624, LR=0.000100
2025-10-06 16:15:47.315 | INFO | main:fit:81 - EPOCH 20000, loss=0.0024, LR=0.000067
2025-10-06 16:15:47.374 | INFO | main:fit:81 - EPOCH 30000, loss=0.0002, LR=0.000050
2025-10-06 16:15:47.434 | INFO | main:fit:81 - EPOCH 40000, loss=0.0000, LR=0.000040
2025-10-06 16:15:47.494 | INFO | main:fit:81 - EPOCH 50000, loss=0.0000, LR=0.000033
2025-10-06 16:15:47.554 | INFO | main:fit:81 - EPOCH 60000, loss=0.0000, LR=0.000029
2025-10-06 16:15:47.614 | INFO | main:fit:81 - EPOCH 70000, loss=0.0000, LR=0.000025
2025-10-06 16:15:47.676 | INFO | main:fit:81 - EPOCH 80000, loss=0.0000, LR=0.000022
2025-10-06 16:15:47.736 | INFO | main:fit:81 - EPOCH 90000, loss=0.0000, LR=0.000020
2025-10-06 16:15:47.798 | INFO | test_main:test_regression_gd:44 - model.weights=array([3.]), model.intercept=np.float64(3.999727042914143)
.
===== 2 passed in 1.01s =====
```

Part. 2, Questions (40%):

1. (10%) Linear models $y = w^T x + b$ have limited fitting power.
 - a. In one sentence, explain why a single linear model is limited.
 - b. Give one concrete task that a single linear model cannot solve, and state why no single hyperplane/affine function solves it.

A single linear model is limited because it can only capture linear relationships between features and the target variable, failing to model complex, non-linear patterns or interactions in the data.

The XOR classification task because the data points (e.g., (0,0) and (1,1) in one class, (0,1) and (1,0) in another) are not linearly separable, meaning no single hyperplane or affine function can correctly classify all points without errors.

2. (15%) Why do we add a regularization term in linear regression? What are the differences between L2 regularization (Ridge) and L1 regularization (Lasso)? Please explain in detail.

We add a regularization term in linear regression to prevent overfitting and improve the model's generalization ability. Regularization penalizes large weights, making the model simpler and less sensitive to noise in the training data.

L2 regularization (Ridge):

- Add a penalty term of the sum of squared weights.
- Shrinks weights towards zero but not forcing to be exactly zero.
- Solution is always unique and smooth.

L1 regularization (Lasso):

- Add a penalty term of the sum of absolute weights.
- Shrinks some weights to exactly zero, performing features selection.
- Solution may not be unique.

3. (15%)
 - What is overfitting? Under what conditions can a model overfit? (List two) How can overfitting be alleviated? (List two)

Overfitting occurs when the model learns the training data too much, including noise and details, which make it poor on new, unseen data.

When the model trained for too many epochs (overtraining), or when the training data is too small or containing lot of noises.

Increase the amount of training data or use data augmentation or do early stopping when the loss stops improving.